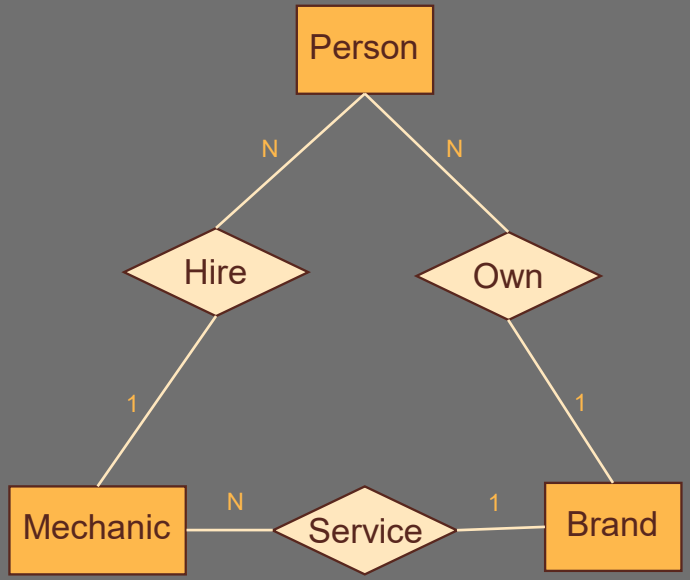
ERD Limitations

Restrictions we can get around

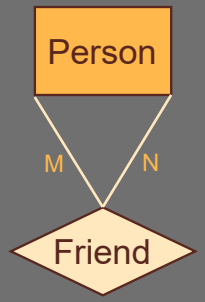
- Restrictions on cardinalities
- No flexibility on relationship keys
- Only unions in classes

Relationship cycles allow inconsistency



- Pam owns only a BMW.
- Pam hires only Mike.
- Mike services only Buicks.

Relationship cycles allow inconsistency



- Pam is friends with Penny.
- Penny is **not** friends with Pam.

- ERD cannot enforce symmetry of “mutual friends”.



Implementation “Fixes”

Can add restrictions via

- User-defined constraints (SQL)
- Triggers (SQL)
- Functions to check conditions (SQL/application)
- CRUD API (SQL/application)

Pragmatic Design Choice

Choose simplest, most-accurate ERD allowing desired semantics.

Impose additional restrictions in implementation.